Guangdong Technion
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GTIIT

Calculus 1 - M
Winter Semester - 2024

Lecture 23 - December 30th

SERIES

Main Example: Geometric Series

$\forall\ -1 < r < 1$, the series is convergent:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{(1-r)} \text{ and } \sum_{n=N_0}^{\infty} r^n = \frac{r^{N_0}}{(1-r)}$$

Example. Compute $\displaystyle\sum_{n=4}^{\infty} \frac{3^n}{2^{2n}}$.

Solution. $\displaystyle\sum_{n=4}^{\infty} \left(\frac{3^n}{2^{2n}}\right) = \sum_{n=4}^{\infty} \left(\frac{3}{4}\right)^n = \left(\frac{3}{4}\right)^4 \sum_{n=4}^{\infty} \left(\frac{3}{4}\right)^{n-4}$

$$= \left(\frac{3}{4}\right)^4 \sum_{k=0}^{\infty} \left(\frac{3}{4}\right)^k = \left(\frac{3}{4}\right)^4 \frac{1}{1-\frac{3}{4}} = \frac{3^4}{4^3} \checkmark$$

Theorem. Integral Test.

Let f be continuous, positive, decreasing on $[1, \infty)$ such that $f(n) = a_n$. Thus, the series $\sum_{k=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x)\,dx$ is convergent.

This means

1. If $\int_1^{\infty} f(x)\,dx$ is convergent, then $\sum_{k=1}^{\infty} a_k$ is convergent, and reciprocally.
2. If $\int_1^{\infty} f(x)\,dx$ is divergent, then $\sum_{k=1}^{\infty} a_k$ is divergent, and reciprocally.

- The starting term may be any $k_0 \in \mathbb{N}$, and so, the interval for the corresponding improper integral is (k_0, ∞) ✓

1. Theorem Let f be a continuous, positive, decreasing function on $[1, \infty)$ such that $f(n) = a_n$.

Denote by $S_n = \sum_{k=1}^{n} a_k$. Suppose that the series is convergent, that is, $\sum_{k=1}^{\infty} a_k = S \in \mathbb{R}$.

Thus, the remainder R_n is bounded by the improper integrals:

$$\boxed{\int_{n+1}^{\infty} f(x)\, dx \leq R_n = S - \sum_{k=1}^{n} a_k \leq \int_{n}^{\infty} f(x)\, dx}$$

Equivalently, the sum of the convergent series is bounded:

$$\boxed{S_n + \int_{n+1}^{\infty} f(x)\, dx \leq S \leq S_n + \int_{n}^{\infty} f(x)\, dx}$$

Remainder of a p - Series: $p > 1$

- If $\sum b_n$ is a p-series, we can estimate its remainder:

$$\boxed{\frac{1}{(p-1)(n+1)^{p-1}} = \int_{n+1}^{\infty} \frac{1}{x^p} dx \leq R_n = S - \sum_{k=1}^{n} \frac{1}{n^p} \leq \int_{n}^{\infty} \frac{1}{x^{p+1}} dx = \frac{1}{(p-1)n^{p-1}}}$$

2. Theorem. The Comparison Test Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with $0 < a_n \leq b_n \; \forall n$. Then

1. If $\sum_{n=1}^{\infty} b_n$ is convergent

   $\sum a_n$ is also convergent.

2. If $\sum_{n=1}^{\infty} a_n$ is divergent then

   $\sum_{n=1}^{\infty} b_n$ is also divergent.

- If we have a series whose terms are smaller than those of a known convergent series, then our series is also convergent.
- If we start with a series whose terms are larger than those of a known divergent series, then it too is divergent.

- $a_n, b_n > 0 \Longrightarrow 0 < \sum a_n$ and $\sum b_n$ are increasing.

- $$0 < a_n \leq b_n \Longrightarrow 0 < \sum_{k=1}^{n} a_k \leq \sum_{k=1}^{n} b_k$$

- Since $\sum_{k=1}^{\infty} b_k = S \in \mathbb{R}$, the partial sums $\sum_{k=1}^{n} a_k$ form a bounded increasing sequence, which is convergent.

- It follows that the sum of the series $\sum_{n=1}^{\infty} a_n$ is less than the sum of the convergent series $\sum_{n=1}^{\infty} b_n \Longrightarrow$ $\boxed{\sum_{n=1}^{\infty} a_n \text{ is convergent:}}$

$$\boxed{\Longrightarrow 0 < \sum_{n=1}^{\infty} a_n \leq \sum_{n=1}^{\infty} b_n < \infty \checkmark}$$

The Comparison Test

- Analogously for the divergence case:

$$\boxed{\sum_{n=1}^{\infty} a_n = \infty \text{ and } \sum_{n=1}^{\infty} a_n \leq \sum_{n=1}^{\infty} b_n \Longrightarrow \sum_{n=1}^{\infty} b_n = \infty \checkmark}$$

3. Example. Test the series $\sum_{n=1}^{\infty} \frac{1}{2^n+n-1}$ for convergence or divergence.

Solution. • Denote by $a_n = \frac{1}{2^n+n-1} \; : \; n \geq 1$

• Let $b_n = \frac{1}{2^n}$.

$$n \geq 1 \Longrightarrow n-1 \geq 0 \Longrightarrow 2^n+n-1 \geq 2^n > 0 \Longrightarrow \frac{1}{2^n} \geq \frac{1}{2^n+n-1} \; : \; n \geq 1$$

• Our given series has smaller terms than those of the geometric series with $r = \frac{1}{2}$, which is convergent, so

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{2^n+n-1} \leq \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2}\frac{1}{1-\frac{1}{2}} = 1}$$

$$\boxed{\Longrightarrow \sum_{n=1}^{\infty} \frac{1}{2^n+n-1} \text{ is convergent } \checkmark}$$

Ex. 4 Test $\sum_{k=1}^{\infty} \frac{\ln(k)}{k}$ for convergence or divergence.

Solution. Compare $a_k = \frac{\ln(k)}{k} > \frac{1}{k} : k \geq 3$:

- $y = \ln(x)$ is increasing, $\ln(e) = 1 \Rightarrow \ln(k) \geq 1 \; \forall k \geq 3$

$$\Rightarrow \boxed{a_k = \frac{\ln(k)}{k} \geq b_k = \frac{1}{k} : k \geq 3}$$

- The harmonic series is divergent by the Integral Test:

$$\int_1^{\infty} \frac{1}{x}\, dx = \lim_{t \to \infty} \left[\ln(t) - \ln(1) \right] = \infty \Longrightarrow \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

- Our series has greater terms than the harmonic series:

$$\boxed{\sum_{n=3}^{\infty} \frac{\ln(n)}{n} \geq \sum_{n=3}^{\infty} \frac{1}{n} = \infty}$$

Limit Comparison Test. Suppose $a_n, b_n > 0 \ \forall n$.

a) If $\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$ then either both series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent or both series are divergent.

b) If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum_{n=1}^{\infty} b_n$ is convergent then the series $\sum_{n=1}^{\infty} a_n$ is also convergent.

c) If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum_{n=1}^{\infty} a_n$ is divergent then the series $\sum_{n=1}^{\infty} b_n$ is also divergent.

Proof of the Limit Comparison Test.

- Let $\epsilon = \frac{c}{2}$ and apply the limit definition.
- Because $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = c > 0$ then there exists N_0 such that

$$n > N_0 \implies \left|\tfrac{a_n}{b_n} - c\right| < \tfrac{c}{2}$$

$$\implies \frac{c}{2} < \frac{a_n}{b_n} < \frac{3c}{2} \quad \boxed{\implies \frac{c}{2} b_n < a_n < \frac{3c}{2} b_n}$$

- So, if $\sum b_n$ converges, so does $\frac{3c}{2}\sum b_n$

$\implies \sum a_n$ converges by the Comparison Test.

- So, if $\sum b_n$ diverges, so does $\frac{c}{2}\sum b_n$

$\implies \sum a_n$ diverges by the Comparison Test ✓

Ex. 6. a) Test the series $\sum_{n=1}^{\infty} \frac{4n^2+\sqrt{3n-1}}{(7n^5-2)}$ for convergence or divergence.

- The leading terms are those of the largest powers:

$$\boxed{a_n = \frac{4n^2+\sqrt{3n-1}}{(7n^5-2)} = \frac{n^2}{n^5}\frac{(4+\sqrt{\frac{3n-1}{n^4}}}{(7-2/n^5)} = \frac{1}{n^3}\frac{4+\sqrt{(3n-1)/n^4}}{(7-2/n^5)}}$$

- Take $b_n = \frac{1}{n^3}$

$$\lim_{n\to\infty}\frac{a_n}{\frac{1}{n^3}} = \lim_{n\to\infty}\frac{n^3}{n^3}\frac{(4+\sqrt{(3n-1)/n^4})}{(7-2/n^5)} = \lim_{n\to\infty}\frac{(4+\sqrt{(3n-1)/n^4})}{(7-2/n^5)}$$

$$= \boxed{\frac{4}{7} = c > 0}$$

The Limit Comparison Test

- The series $\sum b_n = \sum_{n=1}^{\infty} \frac{1}{n^3}$ is a p-series with $p = 3 > 1$,

which is convergent because of the Integral Test:

$$= \int_1^{\infty} \frac{1}{x^3}\, dx = \frac{1}{2} \Longrightarrow \sum_{n=1}^{\infty} \frac{1}{n^3} < \infty$$

Then, the The Limit Comparison Test implies that our series $\sum a_n$ is also convergent:

$$\sum_{n=1}^{\infty} \frac{4n^2 + \sqrt{3n-1}}{(7n^5 - 2)} < \infty \checkmark$$

Ex. 6. b) Test the series $\sum_{n=1}^{\infty} \frac{\sqrt{2n+8}}{\sqrt[4]{3n^5+10}}$ for convergence or divergence.

- The leading terms are those of the largest powers:

$$\boxed{a_n = \frac{(2n+8)^{1/2}}{(3n^5+10)^{1/4}} = \frac{n^{1/2}}{n^{5/4}} \frac{(2+8/n)^{1/2}}{(3+10/n^5)^{1/4}} = \frac{1}{n^{3/4}} \frac{(2+8/n)^{1/2}}{(3+10/n^5)^{1/4}}}$$

- Take $b_n = \frac{1}{n^{3/4}}$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{1}{n^{3/4}} \frac{(2+8/n)^{1/2} \cdot n^{3/4}}{(3+10/n^5)^{1/4}} = \lim_{n\to\infty} \frac{1}{\cancel{n^{3/4}}} \frac{(2+8/n)^{1/2} \cdot \cancel{n^{3/4}}}{(3+10/n^5)^{1/4}}$$

$$= \boxed{\frac{2^{1/2}}{3^{1/4}} = c > 0}$$

The Limit Comparison Test

- The series $\sum b_n = \sum_{n=1}^{\infty} \frac{1}{n^{3/4}}$ is a p-series with $p = 3/4 < 1$, which diverges because of the Integral Test:

$$\boxed{= \int_1^{\infty} \frac{1}{x^{3/4}}\, dx = \infty \Longrightarrow \sum_{n=1}^{+\infty} \frac{1}{n^{3/4}} = \infty}$$

Then, the The Limit Comparison Test implies that our series $\sum a_n$ is also divergent:

$$\boxed{\sum_{n=1}^{\infty} \frac{\sqrt{2n+8}}{\sqrt[4]{3n^5+10}} = \infty \checkmark}$$

- If we have used the Comparison Test to show that $\sum a_n$ converges by comparison with a convergent series $\sum b_n$, then we can estimate the sum $\sum a_n$ by comparing remainders.

  - Consider the reminder of $\sum a_n$:

$$R_n = \sum_{k=1}^{\infty} a_k - \sum_{n=1}^{n} a_k = a_{n+1} + a_{n+2} + \ldots$$

  - Consider the remainder of $\sum b_n$:

$$T_n = \sum_{k=1}^{\infty} b_k - \sum_{n=1}^{n} b_k = b_{n+1} + b_{n+2} + \ldots$$

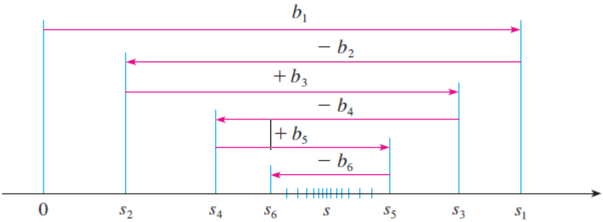
  - Since $\boxed{a_n < b_n \ \forall n}$, then $\boxed{R_n < T_n \ \forall n}$

8. Theorem. Let $b_n > 0$. If the alternating series

$$\sum_{n=1}^{\infty}(-1)^{n-1}b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \ldots$$ verifies

1. $\{b_n\}_n$ is decreasing: $b_{n+1} \leq b_n$
2. $\lim_{n\to\infty} b_n = 0$

then the alternating series $\sum_{n=1}^{\infty}(-1)^{n-1}$ is convergent.



Alternating Series Test: Proof.

1. Consider the partial sums $S_1 = b_1$,

   $S_2 = b_1 - b_2 < S_1, \quad S_3 = S_2 + b_3$ is larger but only a little bit...

2. Continuing in this manner, we see that the partial sums oscillate back and forth.
3. Since $b_n \to 0$, the successive steps are becoming smaller and smaller.
4. The even partial sums $S_2, S_4, S_6, \ldots$ are increasing and the odd partial sums $S_1, S_3, S_5, \ldots$ are decreasing.
   $\Longrightarrow$ both converge to some S

Even Partial Sums are increasing and bounded

- $S_{2n} = S_{2n-2} + (b_{2n-1} - b_{2n}) \geq S_{2n-2}$ since $b_{2n-1} - b_{2n} \geq 0$
  - $S_{2n} = b_1 - (b_2 - b_3)_{>0} - (b_{2n-2} - b_{2n-1})_{>0} - b_{2n} < b_1$

So, the sequence $\{S_n\}_n$ of the even partial sums is increasing and bounded above. It is therefore convergent, there exist $S \in \mathbb{R}$:

$$\lim_{n\to\infty} \sum_{k=1}^{2n} (-1)^k b_k = S$$

- Analogously, it can be shown that the odd partial sums are convergent to the same S:

$$\lim_{n\to\infty} S_{2n+1} = \underbrace{\lim_{n\to\infty} \sum_{k=1}^{2n} (-1)^k b_k}_{\to S} + \underbrace{b_{2n+1}}_{\to 0} = S \checkmark$$

Alternating Series Test: Example.

The alternating harmonic series is convergent:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$$

The general term is decreasing and it tends to 0:

- $0 < \frac{1}{n+1} < \frac{1}{n} \implies b_{n+1} < b_n$
- $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n} = 0$ ✓

So, the theorem implies the convergence of the alternating series ✓.

9. Example. Test the series for convergence or divergence $\displaystyle\sum_{n=7}^{\infty}(-1)^{n+1}\frac{n^2}{(n^3-216)}$

Solution.

1. $6=\sqrt[3]{216}$, so consider $f(x)=\frac{x^2}{x^3-216}$ on $(6,\infty)$,

$$f'(x)=\frac{2x(x^3-216)-3x^2\cdot x^2}{(x^3-216)^2}=\frac{-x^4-432x}{(x^3-216)^2}<0\ \forall x>6$$

$\Longrightarrow f$ decreasing on $(6,\infty)\Longrightarrow b_n$ decreasing for $n\geq 7$

2. $\lim_{n\to\infty} b_n=\lim_{n\to\infty}\frac{n^2}{n^3\left(1-\frac{216}{n^3}\right)}=\lim_{n\to\infty}\frac{1}{n}\frac{1}{\left(1-\frac{216}{n^3}\right)}=0$

Hence, the alternating series $\displaystyle\sum_{n=7}^{\infty}\frac{(-1)^{n+1}n^2}{(n^3-216)}$ is convergent ✓

Alternating Series Estimation Theorem

10. Theorem If $S = \sum_{n=1}^{\infty} (-1)^{n-1} b_n$, where $b_n > 0$, is the sum of an alternating series that satisfies

$$(i)\ b_{n+1} \leq b_n \text{ and } (ii)\ \lim_{n\to\infty} b_n = 0$$

then the remainder is bounded by the next term, that is, $|R_n| = |S - S_n| \leq b_{n+1}$ ✓.

Ex. 11. Find $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ correct to three decimal places.

- We need to find n such that

$$|R_n| \le b_{n+1} = \frac{1}{(n+1)!} \le 0,0001 = 10^{-4} \Longrightarrow 10,000 \le (n+1)!$$

- $8! = 40,320 > 10,000 \Longrightarrow n+1 = 8 \Longrightarrow n = 7.$

$$\boxed{S_7 = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} = \frac{103}{280} = 0.367857\ldots}$$

- $|R_n| \le \frac{1}{8!} \approx 0.0000248 < 10^{-4}$.

- Estimate the series by the partial sum S_7 with the desired accuracy:

$$\boxed{\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} = \frac{1}{e} \approx 0.36787944\ldots}$$

12. Definition. Absolute Convergence.

- A series $\sum_{n=1}^{\infty} a_n$ is called absolutely convergent if the series of the absolute values $\sum_{n=1}^{\infty} |a_n|$ is convergent.

- A series $\sum_{n=1}^{\infty} a_n$ is called conditionally convergent if it is convergent but not absolutely convergent.

- This means that, $\sum_{n=1}^{\infty} a_n$ is conditionally convergent if there exists $L \in \mathbb{R}$ such that

$$\sum_{n=1}^{\infty} a_n = L \text{ but } \sum_{n=1}^{\infty} |a_n| = \infty$$

Example. The alternate harmonic series is only conditionally convergent:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} = L \in \mathbb{R} \text{ but } \sum_{n=1}^{\infty} \frac{1}{n} = +\infty \checkmark$$

Theorem. 13. Absolute Convergence implies Convergence.

If a series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then it is convergent.

The proof is a consequence of the following inequality because $|a_n| = \pm a_n$:

$$0 \le a_n + |a_n| \le 2|a_n|$$

- If $\sum a_n$ is absolutely convergent, then, by definition, $\sum |a_n|$ is convergent, and also $\sum 2|a_n|$ is convergent, therefore, by the Comparison test, $\sum \left(a_n + |a_n|\right)$ is convergent.

$$\implies \sum a_n = \sum \left(a_n + |a_n|\right) - \sum |a_n|$$

is the difference of two convergent sequences, and is therefore convergent ✓

Example of the Theorem. 13. Absolute Convergence implies Convergence.

Ex. 14. Determine whether $\sum_{n=1}^{\infty} \frac{\cos(n)}{n^2}$ is convergent.

Solution. • Apply the theorem to the series of the absolute values:

$$|\cos(n)| \leq 1 \ \forall n \Longrightarrow \frac{|\cos(n)|}{n^2} \leq \frac{1}{n^2}$$

• $\sum_{n=1}^{\infty} \frac{1}{n^2} = L \in \mathbb{R}$ is convergent because of the Integral test.

$$\Longrightarrow 0 \leq \sum_{n=1}^{\infty} \left| \frac{\cos(n)}{n^2} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$\Longrightarrow \sum_{n=1}^{\infty} \frac{\cos(n)}{n^2}$ is absolutely convergent, and then, it is also convergent: $\Longrightarrow \sum_{n=1}^{\infty} \frac{\cos(n)}{n^2} = S \in \mathbb{R}$ ✓

15. Theorem. The Ratio Test.

Let $\sum_{n=1}^{\infty} a_n$ and consider the limit of $\left|\frac{a_{n+1}}{a_n}\right|$ $\forall a_n \neq 0$:

1. If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

2. If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L > 1$ or $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

3. If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L = 1$, then the Test is inconclusive; that is, no conclusion can be drawn about the convergence or divergence of $\sum^{\infty} a_n$.

The Ratio Test.

Ex. 16. Test the convergence or divergence of

$$\sum_{n=1}^{\infty} \frac{n^n}{n!}$$

Solution. $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{(n+1)^{n+1} \cdot n!}{n^n (n+1)!}$

$$= \lim_{n\to\infty} \left(\frac{n+1}{n} \right)^n \frac{(n+1)\cdot n!}{n!(n+1)} = \lim_{n\to\infty} \left(\frac{n+1}{n} \right)^n = e > 1$$

Hence, the given series $\sum_{n=1}^{\infty} \frac{n^n}{n!}$ is divergent ✓

The Root Test. 17. Theorem. The same result as in the theorem 15 is true for the limit $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$.

Let $\sum_{n=1}^{\infty} a_n$ and consider the limit $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$.

1. If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.
2. If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$ or $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.
3. If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L = 1$, the Test is inconclusive: there's no conclusion about the convergence of $\sum_{n=1}^{\infty} a_n$.

Ex. Check the convergence and absolutely convergence of the series $\sum_{n=1}^{\infty}(-1)^n\left(\frac{2n+3}{5n+2}\right)^n$.

Solution. Apply the Root Test to the absolute values terms:

$$L=\lim_{n\to\infty}\sqrt[n]{|a_n|}=\lim_{n\to\infty}\sqrt[n]{\left|(-1)^n\left(\tfrac{2n+3}{5n+2}\right)\right|^n}=\lim_{n\to\infty}\sqrt[n]{\left(\tfrac{2n+3}{5n+2}\right)^n}=\lim_{n\to\infty}\frac{2n+3}{5n+2}=\frac{2}{5}$$

- So, the Root Test implies that $\sum_{n=1}^{\infty}(-1)^n\left(\frac{2n+3}{5n+2}\right)^n$ is absolutely convergent.
- Hence, the series is also convergent. These mean that both, $\sum_{n=1}^{\infty}\left(\frac{2n+3}{5n+2}\right)^n$ and $\sum_{n=1}^{\infty}(-1)^n\left(\frac{2n+3}{5n+2}\right)^n$ are convergent.